ExactFuse: Closed-Form Unfolding with Exact Operators for Sensor-Independent Hyperspectral and Multispectral Image Fusion

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Abstract

Deep unfolding networks for hyperspectral and multispectral image fusion alternate data steps with learned priors, but their blur, downsampling and spectral-response operators are usually learned or zero-padded, so they differ from the operators that produced the data. We study what exact operators bring to such a network. With a separable blur and the true border rule, the data step has a closed form in a product eigenbasis, and the null space of fusion, the part of the image that no observation constrains, is explicit. An exact step keeps the ground truth fixed, an inexact one moves it by an amount that we bound, and blur, spectral response and sub-pixel grid offsets can be estimated from the two observations. Our network, ExactFuse, takes this closed-form step with learned spectral gains, gives its prior inputs in the hyperspectral bands only, so that one network can serve any multispectral sensor, and is trained over a family of observation models whose operators it estimates at test time. Against eleven published networks retrained from their authors' code under one protocol and budget, ExactFuse has the highest PSNR on Chikusei and Pavia Centre, by 0.79 and 0.55 dB, and in 6 of 6 further settings up to $\times 16$, and it reproduces the low-resolution input 17 dB more accurately than the strongest of them. A prior without data steps already beats every baseline, so much of the margin comes from the prior and its training; the exact operators make the result consistent with its inputs, replaceable at test time and robust to changed blurs, responses and noise.

1. Introduction

Hyperspectral cameras record hundreds of narrow bands at a coarse spatial resolution, and multispectral cameras record a few broad bands at a fine one. Fusing the two images gives a high-resolution hyperspectral image (HR-HSI) [28]. Most recent methods are deep networks trained on simulated pairs: convolutional networks [40], transformers [6, 13, 16], state-space models [18] and diffusion models [5].

Deep unfolding networks [8, 11, 30, 31] take a different route. They unroll an iterative solver of the inverse problem, and each stage applies a data step, which reduces the mismatch between the estimate and the two observations, followed by a learned prior. The data step needs the blur, the downsampling and the spectral response of the sensors. In most unfolding networks these operators are learned convolution layers, or fixed but zero-padded, whereas the benchmark data are simulated with a known blur and a different border rule. The network's operator then differs from the one that produced the data, and even the ground truth fails the network's own consistency check. Each network is also trained for one sensor pair and cannot use another operator at test time.

Evaluation is a second problem. Results on the same dataset vary widely between papers because splits, spectral responses, normalizations and metric code differ [28]. PSRT [6], for example, is reported at 44.60 dB PSNR on Chikusei $\times 4$ in [18] and at 50.67 dB in [5]. Numbers copied from different papers therefore say little about which method is better.

This paper asks what exact operators bring to an unfolded fusion network. We write fusion as maximum a posteriori (MAP) estimation with parametric operators, unroll half-quadratic splitting (HQS) [38], and study the resulting data step. None of the fusion networks we compare with uses the border rule of the simulation, is trained over a family of observation models, or works with another multispectral sensor without retraining (the supplementary material). Our contributions are:

- An analysis of the operators in unfolded fusion: with a separable blur and the true border rule, the data step has a closed form and the null space of fusion is explicit (it holds 88% of the unknowns on Chikusei); an exact data step keeps the ground truth fixed, while the displacement caused by an inexact one is bounded independently of the learned weights; and both operators are identifiable from the observation pair (Sec. 3).
- ExactFuse, an unfolded network with exact and replaceable operators, a learned spectral data step that keeps these guarantees, inputs in the hyperspectral bands that do not depend on the multispectral sensor, and training over a family of observation models with the operators estimated at test time (Sec. 4).
- An equal-budget benchmark of eleven published net-

works, retrained from their authors' code, and a null-space split of the error that shows where each method fails (Sec. 5). It also shows where ExactFuse's margin comes from: much of it from the prior and its training, which keep a lead under equal recipes, while the exact operators make the result consistent with its inputs and matter most when the operator changes.

2. Related work

Deep fusion networks map the two observations directly to the HR-HSI with convolutions [40], transformers [6, 9, 13, 16, 32], general fusion backbones [19], state-space models [18] or diffusion [5], without using the observation model. SSA [15] slices its convolution kernels to the number of bands and reconstructs at arbitrary scales, so one model serves several sensors; it does not model their responses.

Unfolding and operator knowledge. Algorithm unrolling [17] turns the iterations of a solver into network stages. MHF-Net [30], MoG-DCN [8] and DHIF-Net [11] unfold fusion solvers with learned or approximate operators, and later networks strengthen the learned prior [31] or the proximal update [35]; IR&ArF [22] unfolds the fusion of unregistered images with a learned registration. In single-image super-resolution, USRNet [38] gives the blur kernel to a closed-form data step under periodic borders and predicts its step sizes from the noise level, and blind methods train on randomized degradations [39]. We carry this idea over to fusion, where two operators and the border rule must match the data, and analyze what it gives and costs.

Closed forms and separable operators. FUSE [29] solves the quadratic fusion problem through a Sylvester equation under periodic borders. Tensor methods [12, 14, 21] model the spatial degradation as separable matrices to prove recoverability of low-rank models. Our closed form holds for the reflect border of the simulation and is used as a differentiable layer whose operators change from image to image. Learned filters in a data step resemble deep Wiener deconvolution [7]; we bound ours so that the step stays non-expansive.

Range and null space. Splitting a reconstruction into a part determined by the measurements and a null-space part is the basis of range–null-space decomposition networks [4]. In pansharpening, back-projecting a fused image onto its observations improves its consistency [26]. With exact separable operators this projection has a closed form (Theorem 1); we use the explicit null space to analyze the errors of existing methods. That an adjoint that does not match the data biases an iterative reconstruction is known from tomography [37].

3. Exact operators in unfolded fusion

3.1. Observation model and unfolding

Let $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ be the HR-HSI. The two observations are

$$\mathbf{Y}_{\mathrm{H}} = \mathcal{D}_\theta \mathbf{X} + \mathbf{N}_{\mathrm{H}}, \qquad \mathbf{Y}_{\mathrm{M}} = \mathcal{R}_\phi \mathbf{X} + \mathbf{N}_{\mathrm{M}}, \tag{1}$$

where $\mathcal{D}_\theta$ applies to every band reflect padding, a Gaussian blur of width θ and decimation by s starting at index zero, so $\mathbf{Y}_{\mathrm{H}} \in \mathbb{R}^{B\times h\times w}$ with $h = H/s$, $w = W/s$; $\mathcal{R}_\phi$ applies to every pixel the M spectral responses of the multispectral sensor, shifted along wavelength by ϕ nm; and $\mathbf{N}_{\mathrm{H}}, \mathbf{N}_{\mathrm{M}}$ are white Gaussian noise with standard deviations $\varsigma_{\mathrm{H}}, \varsigma_{\mathrm{M}}$. The benchmarks use $s = 4$, a 7×7 blur with $\theta = \sigma_0 = 2$, $\phi = 0$ and no noise [5]. We implement $\mathcal{D}_\theta$ and its exact adjoint, in which the adjoint of reflect padding adds the padded margins back onto the pixels they were copied from (the supplementary material); the adjoint test $\langle \mathcal{D}\mathbf{X}, \mathbf{Y}\rangle = \langle \mathbf{X}, \mathcal{D}^\top \mathbf{Y}\rangle$ holds to 10^{-13} in double precision.

With a prior $p(\mathbf{X}) \propto \exp(-\lambda\Phi(\mathbf{X}))$, the MAP estimate minimizes $\frac{1}{2\varsigma_{\mathrm{H}}^2}\|\mathcal{D}_\theta\mathbf{X}-\mathbf{Y}_{\mathrm{H}}\|^2 + \frac{1}{2\varsigma_{\mathrm{M}}^2}\|\mathcal{R}_\phi\mathbf{X}-\mathbf{Y}_{\mathrm{M}}\|^2 + \lambda\Phi(\mathbf{X})$. HQS couples $\mathbf{X}$ to an auxiliary image $\mathbf{Z}$ with a penalty $\frac{\mu_k}{2}\|\mathbf{Z}-\mathbf{X}\|^2$ and alternates a data step and a prior step, $\mathbf{X}_k = \mathrm{prox}_{(\lambda/\mu_k)\Phi}(\mathbf{Z}_k)$. With $\alpha_k = 1/(\mu_k\varsigma_{\mathrm{H}}^2)$ and $\beta_k = 1/(\mu_k\varsigma_{\mathrm{M}}^2)$ the data step is the linear system

$$(\mathbf{I}+\alpha_k\mathcal{D}_\theta^\top\mathcal{D}_\theta+\beta_k\mathcal{R}_\phi^\top\mathcal{R}_\phi)\mathbf{Z}_k = \mathbf{X}_{k-1}+\alpha_k\mathcal{D}_\theta^\top\mathbf{Y}_{\mathrm{H}}+\beta_k\mathcal{R}_\phi^\top\mathbf{Y}_{\mathrm{M}}, \tag{2}$$

and one gradient step from $\mathbf{X}_{k-1}$ is its linearization,

$$\mathbf{Z}_k \approx \mathbf{X}_{k-1}-\alpha_k\mathcal{D}_\theta^\top(\mathcal{D}_\theta\mathbf{X}_{k-1}-\mathbf{Y}_{\mathrm{H}})-\beta_k\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{X}_{k-1}-\mathbf{Y}_{\mathrm{M}}). \tag{3}$$

An unfolded network replaces the proximal map by a learned module. The derivation already says two things about such a network: the ratio $\beta_k/\alpha_k = \varsigma_{\mathrm{H}}^2/\varsigma_{\mathrm{M}}^2$ should follow the noise of the two observations, and the operators enter only through $\mathcal{D}_\theta$, $\mathcal{R}_\phi$ and their adjoints, so they can be replaced at test time without retraining.

3.2. Separable structure and the fusion null space

A Gaussian blur aligned with the image axes is separable, and padding and decimation act on rows and columns independently. The operators therefore factor exactly, border included:

$$\mathcal{D}_\theta\mathbf{X} = \mathbf{X}\times_2\mathbf{D}_{\mathrm{r}}\times_3\mathbf{D}_{\mathrm{c}}, \qquad \mathcal{R}_\phi\mathbf{X} = \mathbf{X}\times_1\mathbf{r}_\phi, \tag{4}$$

where $\times_n$ is the mode-n product, $\mathbf{D}_{\mathrm{r}} \in \mathbb{R}^{h\times H}$ and $\mathbf{D}_{\mathrm{c}} \in \mathbb{R}^{w\times W}$ pad, blur and decimate one axis, and $\mathbf{r}_\phi \in \mathbb{R}^{M\times B}$ holds the spectral responses.

Theorem 1 (Closed-form data step and null space)**.** *Let* $\mathbf{D}_{\mathrm{r}}^\top\mathbf{D}_{\mathrm{r}} = \mathbf{U}_{\mathrm{r}}\mathrm{diag}(\boldsymbol{\lambda}^{\mathrm{r}})\mathbf{U}_{\mathrm{r}}^\top$, $\mathbf{D}_{\mathrm{c}}^\top\mathbf{D}_{\mathrm{c}} = \mathbf{U}_{\mathrm{c}}\mathrm{diag}(\boldsymbol{\lambda}^{\mathrm{c}})\mathbf{U}_{\mathrm{c}}^\top$ *and* $\mathbf{r}_\phi^\top\mathbf{r}_\phi = \mathbf{V}\mathrm{diag}(\boldsymbol{\nu})\mathbf{V}^\top$, *and let* $\mathcal{Q}(\cdot) = (\cdot)\times_1\mathbf{V}\times_2\mathbf{U}_{\mathrm{r}}\times_3\mathbf{U}_{\mathrm{c}}$.

(a) $\mathcal{D}_\theta^\top \mathcal{D}_\theta$ and $\mathcal{R}_\phi^\top \mathcal{R}_\phi$ are diagonal in the orthonormal basis $\mathcal{Q}$, with eigenvalues $\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}}$ and ν_b. (b) The solution of (2) is

$$\mathbf{Z}_k = \mathcal{Q}\Big(\mathcal{Q}^{-1}(\mathbf{C}_k) \oslash \big[1 + \alpha_k \lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} + \beta_k \nu_b\big]_{bij}\Big), \quad (5)$$

with $\mathbf{C}_k$ the right-hand side of (2) and $\oslash$ element-wise division. (c) The joint null space $\mathcal{N}(\mathcal{D}_\theta) \cap \mathcal{N}(\mathcal{R}_\phi)$ is spanned by the basis tensors with $\nu_b = 0$ and $\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} = 0$. If $\mathbf{r}_\phi$, $\mathbf{D}_{\mathrm{r}}$ and $\mathbf{D}_{\mathrm{c}}$ have full row rank, its dimension is $(B-M)(HW-hw)$, a fraction $(1 - M/B)(1 - 1/s^2)$ of the unknowns.

The proof (the supplementary material) uses the mixed-product rule of the Kronecker product: every factor of $\mathcal{Q}$ diagonalizes one mode. The step (5) costs three mode products in each direction, $O(BHW(B+H+W))$ operations, and needs no iteration. FUSE [29] solves the same system with Fourier transforms, which assumes periodic borders; (5) holds for the border rule of the simulation, at the cost of eigendecompositions of an $H \times H$, a $W \times W$ and a $B \times B$ matrix per operator. Part (c) states what any fusion method must infer from a prior: the high spatial frequencies of the spectral directions that the multispectral sensor does not see. This is 88% of the unknowns on Chikusei and 90% on Pavia Centre at $\times 4$. Without the HR-MSI the null space would hold all high spatial frequencies, $1 - 1/s^2$ of the unknowns, as in single-image super-resolution. The HR-MSI observes them in only M spectral directions, but spatial detail is so strongly shared across bands that these directions carry 84% of their energy on Chikusei and 77% on Pavia Centre: the minimum-norm image consistent with both observations reaches 41.3 dB on Chikusei, against 33.5 dB for the LR-HSI alone. Every other component is fixed by the observations, which lets us split the error of any method into a part that contradicts the observations and a part in the null space (Sec. 5.3).

3.3. The cost of an inexact operator

Stack the data terms as $\mathbf{A} = [\mathcal{D}_\theta; \mathcal{R}_\phi]$, $\mathbf{Y} = [\mathbf{Y}_{\mathrm{H}}; \mathbf{Y}_{\mathrm{M}}]$ and $\mathbf{W}_k = \mathrm{diag}(\alpha_k \mathbf{I}, \beta_k \mathbf{I})$, and let $\tilde{\mathbf{A}}$ be the operator a network actually uses: one with a zero border, a learned one, or one with wrong parameters. Write $T(\mathbf{V}) = (\mathbf{I} + \tilde{\mathbf{A}}^\top \mathbf{W}_k \tilde{\mathbf{A}})^{-1}(\mathbf{V} + \tilde{\mathbf{A}}^\top \mathbf{W}_k \mathbf{Y})$ for the exact data step and $G(\mathbf{V}) = \mathbf{V} - \tilde{\mathbf{A}}^\top \mathbf{W}_k(\tilde{\mathbf{A}}\mathbf{V} - \mathbf{Y})$ for its linearization.

Theorem 2 (Fixed point and operator mismatch)**.** *Let $\mathbf{Y} = \mathbf{A}\mathbf{X}^*$. (a) If $\tilde{\mathbf{A}} = \mathbf{A}$, then $T(\mathbf{X}^*) = G(\mathbf{X}^*) = \mathbf{X}^*$ for all $\alpha_k, \beta_k > 0$. (b) For any $\tilde{\mathbf{A}}$, with $\mathbf{e} = \mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$,*

$$\|T(\mathbf{X}^*) - \mathbf{X}^*\| \le \tfrac{1}{2}\|\mathbf{e}\|, \qquad \|G(\mathbf{X}^*) - \mathbf{X}^*\| \le \|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\|\,\|\mathbf{e}\|. \quad (6)$$

(c) If stage k computes $\mathbf{X}_k = (\mathcal{I} + \mathcal{P}_k)(T_k(\mathbf{X}_{k-1}))$ with a residual prior that satisfies $\mathcal{P}_k(\mathbf{X}^*) = \mathbf{0}$ and makes $\mathcal{I} + \mathcal{P}_k$ L_k-Lipschitz, then a K-stage network started at $\mathbf{X}^*$ returns $\|\mathbf{X}_K - \mathbf{X}^*\| \le \frac{1}{2}\sum_{k=1}^{K}\big(\prod_{j=k}^{K} L_j\big)\|\mathbf{e}_k\|$, where $\mathbf{e}_k$ is $\mathbf{e}$ for the weights $\mathbf{W}_k$ of stage k.

The bounds follow from the singular values of $(\mathbf{I} + \mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ with $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$, which are $b_i/(1+b_i^2) \le \frac{1}{2}$, and from the nonexpansiveness of T_k (the supplementary material). With the exact operator the ground truth is a fixed point of every data step, and the priors only have to remove reconstruction error. With an inexact one every data step moves the ground truth, and the priors must undo this displacement as well; since it depends on the image, no fixed correction removes it. The exact proximal step bounds the displacement whatever the learned weights are, whereas the gradient step does not, because training can make $\|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\|$ large. The theorem also explains why an exact operator can be swapped: for data generated by $\mathbf{A}'$, giving the network $\tilde{\mathbf{A}} = \mathbf{A}'$ restores (a), which a learned operator cannot do.

Corollary 1 (Border rule)**.** *If $\tilde{\mathbf{A}}$ differs from $\mathbf{A}$ only in the border rule of $\mathcal{D}_\theta$, then $(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$ is nonzero only on the low-resolution pixels whose kernel support reaches the padding, i.e., within $\lceil p/s \rceil$ pixels of the border for a $(2p+1) \times (2p+1)$ kernel.*

For our benchmarks only the first low-resolution row and column are affected: 3% of a 64×64 test image, but 12% of the 16×16 patches on which networks are trained.

3.4. Estimating the operators

Theorem 3 (Identifiability)**.** *Let $\mathcal{D}_\theta$ act identically on every band and $\mathcal{R}_\phi$ identically on every pixel. (a) $\mathcal{D}_\theta\mathcal{R}_{\phi'} = \mathcal{R}_{\phi'}\mathcal{D}_\theta$, so for noise-free data generated with (θ^*, ϕ^*) the criterion*

$$J(\theta, \phi) = \|\mathcal{D}_\theta \mathbf{Y}_{\mathrm{M}} - \mathcal{R}_\phi \mathbf{Y}_{\mathrm{H}}\|^2 \quad (7)$$

vanishes at (θ^*, ϕ^*). (b) On a finite set $\mathcal{G}$ of candidate parameters on which $\theta \mapsto \mathcal{D}_\theta$ and $\phi \mapsto \mathbf{r}_\phi$ are injective, and if no column of $\mathbf{r}_\phi$ vanishes, (θ^*, ϕ^*) is the unique zero of J on $\mathcal{G}$ for almost every $\mathbf{X}$.

The parameter θ may collect several parameters of the separable family, such as the row and column blur widths and the sub-pixel grid offsets of Sec. 5.5; the theorem only needs $\theta \mapsto \mathcal{D}_\theta$ to be injective on $\mathcal{G}$. Part (a) is the relation HySure [24] uses to estimate the responses of a model-based method; part (b) shows that a grid search on J cannot be fooled by a wrong pair except for a set of images of measure zero (the supplementary material). We minimize J on a grid with steps of $0.025\,\sigma_0$ and 1 nm. Noise biases the blur estimate upward, less as the SNR grows, and the family must contain the true operator; Sec. 5.5 tests what happens when it does not.

4. ExactFuse

Stages. ExactFuse (Fig. 1) unrolls HQS for $K = 3$ stages from the bicubic upsampling $\mathbf{X}_0$ of $\mathbf{Y}_{\mathrm{H}}$. Stage k first takes

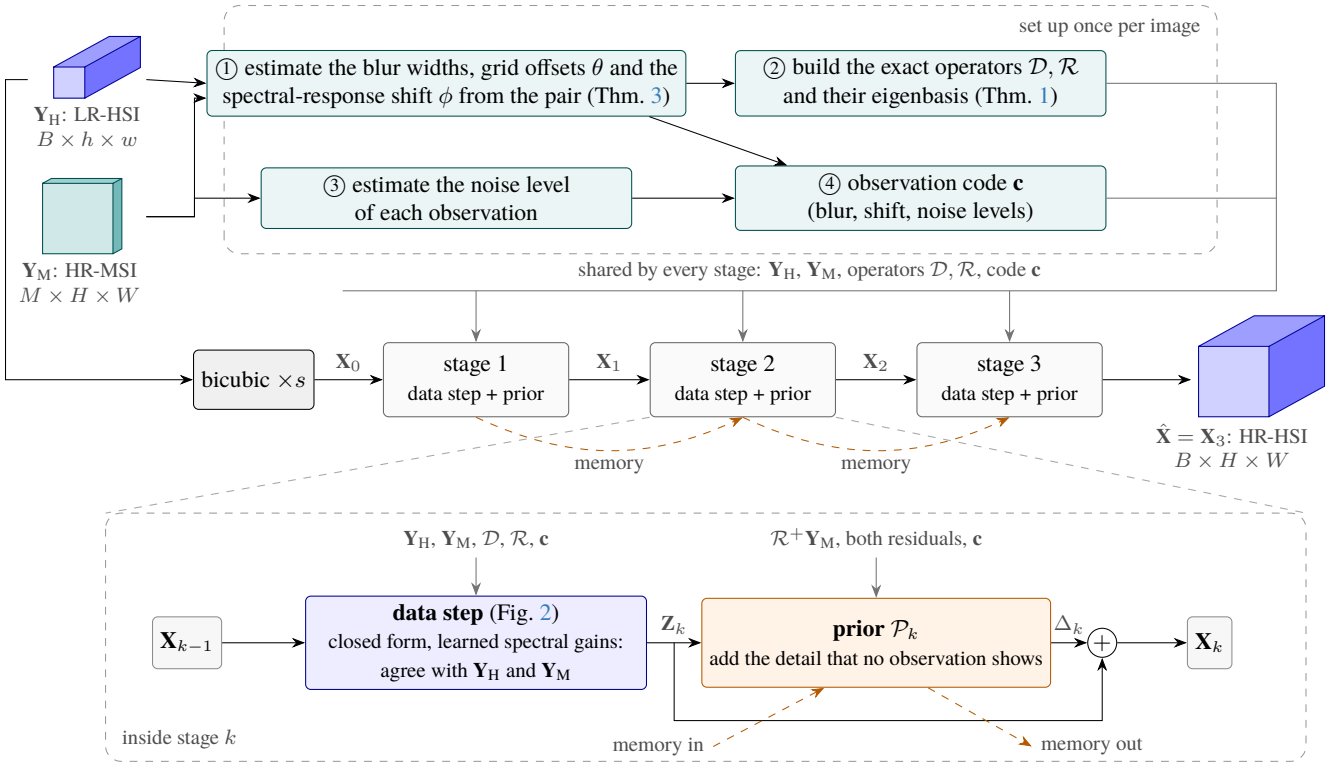


Figure 1. ExactFuse, the HQS iteration of Section 3.1 unrolled for three stages. Top: the observation model is set up once per image: the operator parameters are given or estimated from the observation pair (Theorem 3), the exact operators and their eigenbasis are built (Theorem 1), and the noise levels form the observation code **c**. Middle: every stage refines the estimate and passes its decoder features to the next (memory). Bottom: inside a stage, a data step makes the estimate agree with both observations, and a learned prior adds the detail that the observations cannot provide.

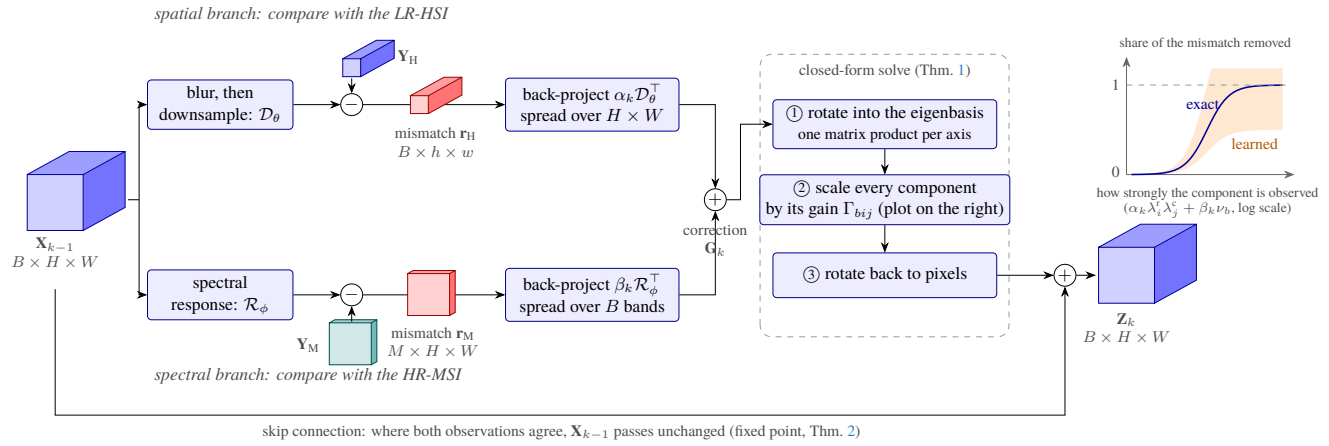


Figure 2. The data step of stage k, the exact solution of (2). Two physics branches compare the estimate with the observations: the spatial branch blurs and downsamples it and compares it with the LR-HSI, the spectral branch applies the multispectral responses and compares it with the HR-MSI; both mismatches are spread back to the full cube by the exact adjoints. The closed-form solve rotates this correction into the eigenbasis of the separable operators, removes in each component a share $\Gamma_{bij}(\alpha_k\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} + \beta_k\nu_b)$ of the mismatch that grows with how strongly the observations see that component (blue: the exact step (5); orange: the range allowed to the learned gains of Proposition 1), and rotates back. Components that neither observation sees (the null space) are left to the prior.

a data step with the exact operators: the learned spectral step (9) below, or, for non-separable operators and in the base model of Sec. 5.4, the linearized step

$$\begin{aligned}\mathbf{Z}_k = \mathbf{X}_{k-1} &- s^2\alpha_k\mathcal{D}_\theta^\top(\mathcal{D}_\theta\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{H}) \\ &- \beta_k\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{M})\end{aligned} \tag{8}$$

The step sizes $\alpha_k = \mathrm{softplus}(a_k)$ and $\beta_k = \mathrm{softplus}(b_k)$ are learned per stage, and the factor s^2 compensates the $1/s^2$ gain of $\mathcal{D}^\top\mathcal{D}$ on smooth images. The prior step is a learned residual, $\mathbf{X}_k = \mathbf{Z}_k + \Delta_k$ with $(\Delta_k, \mathbf{F}_k) = \mathcal{P}_k([\mathbf{Z}_k, \mathcal{R}_\phi^+\mathbf{Y}_\mathrm{M}, s^2\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H}), \frac{B}{M}\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{Z}_k - \mathbf{Y}_\mathrm{M})], \mathbf{F}_{k-1})$. Here $\mathcal{R}_\phi^+\mathbf{Y}_\mathrm{M}$ are the minimum-norm spectra in the B hyperspectral bands that reproduce the HR-MSI ($\mathcal{R}_\phi\mathcal{R}_\phi^+ = \mathbf{I}$, so they determine $\mathbf{Y}_\mathrm{M}$ exactly), and the last two inputs tell the prior where the estimate still disagrees with each observation. Every input lives in the hyperspectral bands, so the prior does not depend on the number or the responses of the multispectral bands. $\mathbf{F}_{k-1}$ are the decoder features of the previous stage (cross-stage memory [25]). The last layer of every prior starts at zero, so training starts from plain HQS, for which the condition $\mathcal{P}_k(\mathbf{X}^*) = \mathbf{0}$ of Theorem 2(c) holds. All operator applications run in 32-bit precision.

Learned spectral data step. In the basis of Theorem 1 the exact step multiplies each component of the data residual by $1/(1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b)$, the filter of a white Gaussian prior. Images are not white and noise does not affect all components equally, so we let two small networks modulate these gains:

$$\begin{aligned}\mathbf{Z}_k &= \mathbf{X}_{k-1} + \mathcal{Q}\big(\mathcal{Q}^{-1}(\mathbf{G}_k) \odot \mathbf{\Gamma}_k\big), \\ \Gamma_{k,bij} &= \frac{\exp\big(\ln 2\, \tanh(u_k(\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}) + v_k(\nu_b))\big)}{1 + \alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b},\end{aligned} \tag{9}$$

where $\mathbf{G}_k = \alpha_k\mathcal{D}_\theta^\top(\mathbf{Y}_\mathrm{H} - \mathcal{D}_\theta\mathbf{X}_{k-1}) + \beta_k\mathcal{R}_\phi^\top(\mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi\mathbf{X}_{k-1})$ and u_k, v_k are two-layer perceptrons of the log-eigenvalues (and of the observation code below), initialized to zero so that training starts from (5). The gain factors over space and spectrum, so the perceptrons run $HW + B$ times per stage, not BHW times.

Proposition 1. *For any u_k, v_k: (a) with exact operators and noise-free data, $\mathbf{X}^*$ is a fixed point of (9); (b) the step is nonexpansive; (c) with an inexact operator it moves $\mathbf{X}^*$ by at most $\|\mathbf{e}\|$, twice the bound (6) of the exact step.*

The gain lies in $(\frac{1}{2}, 2)$ times the exact one, which keeps every factor of the step in $(-1, 1]$ (the supplementary material). The learned step can thus change how much the network trusts each observed component, but it cannot move a correct estimate or amplify errors.

Prior. Each prior is a three-level U-Net [23] (widths 48, 96, 192) of Restormer blocks [36], whose transposed attention forms a channel-by-channel attention map with a cost linear in the number of pixels. ExactFuse holds 10.5 M parameters for Chikusei, almost all in the three priors. Their design is not our contribution, and Sec. 5.4 shows that a convolutional mixer works about as well.

Training over a family of observation models. A network trained with one operator pair on noise-free data adapts its priors to that blur, that response and no noise. Since the operators enter ExactFuse only through its data steps, we can instead draw for every batch a blur width $\sigma \sim \mathcal{U}[0.5, 2]\sigma_0$, a spectral shift $\delta \sim \mathcal{U}[-12, 12]$ nm and, with probability 0.5, noise at an SNR from $\mathcal{U}[30, 50]$ dB, simulate the batch with $(\mathcal{D}_\sigma, \mathcal{R}_\delta)$, and use the same operators in the data steps. At test time the operators are given or estimated with (7). As HQS suggests, we also condition on the noise: a code $\mathbf{c} = (\sigma/\sigma_0 - 1, \delta/12, \log_{10} n_\mathrm{H} + 2, \log_{10} n_\mathrm{M} + 2)$, with the relative noise levels n estimated from each observation by the median absolute deviation of its second differences, sets the step sizes $\alpha_k = \mathrm{softplus}(a_k + \mathbf{u}_k^\top\mathbf{c})$, $\beta_k = \mathrm{softplus}(b_k + \mathbf{v}_k^\top\mathbf{c})$ and modulates the prior features (FiLM [20]), similar to the hyper-parameter module of USRNet [38].

Loss. We train with $\|\mathbf{X}_K - \mathbf{X}\|_1 + 0.1\sum_{k<K}\|\mathbf{X}_k - \mathbf{X}\|_1 + 0.05\,\mathrm{SAM}(\mathbf{X}_K, \mathbf{X}) + 0.1\,(1 - \mathrm{SSIM}(\mathbf{X}_K, \mathbf{X}))$ and AdamW, and test the exponential moving average of the weights.

5. Experiments

5.1. Setup

Data. Chikusei [34] has 128 bands (363–1018 nm) and is paired with the eight WorldView-2 responses; Pavia Centre [10] has 92 bands and is paired with the four IKONOS responses. We use the split sizes of [5] (8 and 4 test images of 256×256 pixels) with fixed, disjoint locations given in the supplementary material. CAVE [33] (20 training and 12 test scenes) and Harvard [3] (30 and 20 images) have 31 bands and are paired with a Nikon RGB response. All pairs follow Wald's protocol [27]: both observations are simulated from the reference with (1), every training crop on its own. At $\times8$ and $\times16$ the blur is a $(2s-1)\times(2s-1)$ Gaussian with $\sigma = s/2$.

Baselines and budget. We compare with GSA [1] and eleven networks: SSR-NET [40], PSRT [6], DCT [16], MIMFormer [13], DHIF-Net [11], U2Net [19], MIMO-SST [9], OTPNet [35], SMGU-Net [31], CLSNet [32] and CaFormer [2]. Each is built from its authors' code; only the numbers of bands and the scale factor are changed (the supplementary material lists the commits). All methods, Ex-

actFuse included, are trained for 4 h on one Tesla T4 GPU with the same patches (64×64), batch size (8), augmentation, EMA weights and validation-only model selection; SSR-NET, which converges sooner, gets 1.5 h. The published networks keep their published learning rates with an L1 loss. The metrics are computed by the same code for all methods: PSNR (peak equal to the dataset maximum), SSIM, the spectral angle SAM (degrees) and ERGAS; Q2n, CC, SCC and RMSE are in the supplementary material.

5.2. Comparison

Table 1 shows the $\times 4$ results. ExactFuse reaches 58.05 dB on Chikusei and 50.63 dB on Pavia Centre, 0.79 and 0.55 dB above the best retrained network, and it has the lowest SAM and ERGAS on both datasets. Over three seeds, the PSNR of ExactFuse and of MIMFormer varies by 0.01 and 0.18 dB on Chikusei, and the worst ExactFuse seed is still 1.09 dB above the best MIMFormer seed. Pooled over the 76 test images of all 8 settings below, ExactFuse is more accurate than the strongest retrained network of the setting on 75 of them (one-sided Wilcoxon signed-rank test, $p = 1.9 \times 10^{-14}$). Table 3 extends the comparison to CAVE and Harvard, whose test scenes are unseen in training, and to $\times 8$ and $\times 16$: ExactFuse has the highest PSNR in 6 of the 6 settings, by 0.21–2.29 dB over the strongest network run in each.

A fixed budget raises two questions. First, does our pipeline weaken the baselines? It does not: the retrained networks match or exceed the values reported for them, e.g., MIMFormer reaches 56.96 dB on Chikusei against 53.36 dB in [5], and their authors' own training recipes do not improve them (the supplementary material). Second, does the budget favor fast networks? Partly: DCT completes only 4,800 iterations in 4 h on Pavia Centre, against 33,850 for ExactFuse. With 8 h of training, ExactFuse and MIMFormer reach 58.18 and 57.37 dB on Chikusei; the gap narrows from 1.10 to 0.81 dB but remains, although ExactFuse completes 63,850 iterations against 186,800.

For context, the supplementary material lists ExactFuse trained for 8 h next to the 17 methods tabulated in [5]; that table cannot rank methods, since our retrained baselines exceed their reported values by up to 3.6 dB, and its strongest entry, a diffusion model, has no public code.

5.3. Where the margin comes from

We reduce ExactFuse step by step (the supplementary material). A single U-Net prior on the bicubic image and the HR-MSI, without data steps or stages, already reaches 57.67 dB on Chikusei and 50.39 dB on Pavia Centre, above every retrained network. Most of the margin therefore comes from this residual multiscale prior and its training recipe, not from the transformer blocks, and the exact data steps add 0.38 and 0.23 dB on top. The recipe is not the whole story: with the same recipe for both, ExactFuse leads MIMFormer by 0.53 and 0.40 dB under ours and by 0.72 and 0.44 dB under the baselines'.

The null space of Theorem 1(c) gives a second view: we split the error of every method into its part in the observed subspace $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, which contradicts the observations, and its null-space part, which only a prior can reduce. Bicubic interpolation and GSA err mostly in the observed subspace (only 8.6% and 13.7% of their squared error lies in the null space on Chikusei). ExactFuse's observed error (column Obs. of Tab. 1) is 27 times smaller than that of the best retrained network on Chikusei (MIMO-SST) and 9 times on Pavia Centre. Removing MIMFormer's observed error alone would raise its PSNR by 0.43 dB on Chikusei; the remaining 0.67 dB of ExactFuse's margin come from a better null-space estimate. The exact data steps account for the first part, the prior for the second.

5.4. What the exact operator does

Table 4 retrains variants from scratch with the same budget. Its top block goes from the base model to ExactFuse. The closed-form step with learned gains and the HR-MSI residual input each raise the agreement with the LR-HSI (89.4 and 86.5 against 83.9 dB on Chikusei) at nearly equal PSNR; combined, with every input in the hyperspectral bands, they change the PSNR by +0.10 and 0.00 dB (Chikusei, Pavia Centre) and raise the agreement to 93.4 dB, with fewer iterations in the budget (30,300 against 35,750). A learned gate that confines the prior to the null space lowers the PSNR (-0.08 and -0.04 dB), and so does, on Pavia Centre, a branch that transfers the HR-MSI's detail to all bands (-0.08 dB); ExactFuse uses neither.

Without data steps ExactFuse loses 0.63 and 0.33 dB and its agreement with the LR-HSI falls from 93.4 to 67.5 dB; a gradient step instead of the closed form keeps the PSNR but lowers the agreement to 86.7 dB. In the base model, a zero border or a learned blur and response cost 0.22 and 0.22 dB on Chikusei and reproduce the LR-HSI about 3 dB worse, as Theorem 2 predicts; over three seeds the zero border costs 0.22 and 0.08 dB, against a seed spread of 0.004 and 0.02 dB.

The difference becomes large when the blur changes (the supplementary material): given the true blur, the exact base model loses at most 0.25 dB on Chikusei, the zero-border model 3.41 dB, although its operator differs only at the image border, and the learned operator, which cannot be replaced, 3.45 dB. A convolutional mixer instead of attention changes the PSNR by only +0.07 and -0.02 dB (Chikusei, Pavia Centre).

5.5. One network for a family of observation models

Table 2 tests networks trained over the operator family on conditions that leave the training protocol. Within the family the estimate of Theorem 3 is accurate: over all Chikusei test images and conditions, 30 dB SNR included, its largest

Table 1. Equal-budget comparison at $\times 4$. Every network is retrained by us from its authors' code with the same data, patches, batch size, EMA weights, validation-only model selection and 4 h on one T4 GPU (single pass). SSIM with the data range set to each image's own maximum. Obs.: mean squared error in the observed subspace ($\times 10^{-6}$), the part of the error that contradicts the observations (Sec. 5.3). Parameters for 128 bands. Best learned method in bold.

		Chikusei $\times 4$					Pavia Centre $\times 4$				
Method	Params (M)	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Obs.↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Obs.↓
Bicubic	–	33.63	0.7404	4.406	9.011	431	25.52	0.6282	6.554	9.852	2746
GSA [1]	–	43.20	0.9778	2.730	3.279	46.9	34.78	0.9438	5.229	3.494	370
SSR-NET [40]	0.44	55.55	0.9958	0.954	1.707	0.703	48.45	0.9954	2.443	0.758	2.049
PSRT [6]	0.30	53.99	0.9948	1.057	1.794	1.645	47.87	0.9958	2.212	0.801	2.483
DCT [16]	8.46	53.18	0.9945	1.133	2.066	2.240	46.94	0.9953	2.442	0.888	4.312
MIMFormer [13]	16.46	56.96	0.9964	0.801	1.488	0.186	49.63	0.9967	1.982	0.674	0.430
DHIF-Net [11]	48.67	54.39	0.9953	1.020	1.660	1.546	47.67	0.9956	2.295	0.840	3.980
U2Net [19]	3.06	53.35	0.9943	1.076	1.948	–	47.29	0.9955	2.327	0.858	–
MIMO-SST [9]	5.32	57.27	0.9965	0.780	1.419	0.066	49.69	0.9967	1.961	0.670	0.257
OTPNet [35]	2.51	57.06	0.9965	0.786	1.448	0.218	50.08	0.9968	1.951	0.653	0.677
SMGU-Net [31]	9.50	55.14	0.9957	0.949	1.705	0.842	48.03	0.9958	2.331	0.783	2.293
CLSNet [32]	4.17	56.65	0.9963	0.811	1.514	0.242	49.50	0.9967	1.993	0.684	0.639
CaFormer [2]	7.83	53.56	0.9949	1.054	2.138	2.095	48.43	0.9961	2.145	0.757	1.713
ExactFuse	10.45	**58.05**	**0.9969**	**0.724**	**1.334**	**0.002**	**50.63**	**0.9970**	**1.892**	**0.619**	**0.028**

Table 2. One network for a family of observation models: mean PSNR (dB) over groups of test conditions. Randomized: blur width (0.5–$2\sigma_0$), SRF shift (± 12 nm) and noise (30–50 dB) drawn anew for every training batch. The operator is the training one (MIMFormer, randomized data), or estimated from the test pair with (7); the last row uses the true operator (reference, not marked). †Anisotropic and defocus blurs, outside the family. §LR-HSI sampled on a grid offset by 0.5–1 HR pixel (output on the HR-MSI grid), alone and with an axis-aligned anisotropic blur. ‡HR-MSI shifted by 0.5–1 pixel against the reference. Best per column in bold.

	Chikusei $\times 4$								Pavia Centre $\times 4$							
Model	Nom.	Blur	SRF	Noise	All	Kernel†	Offset§	Misreg.‡	Nom.	Blur	SRF	Noise	All	Kernel†	Offset§	Misreg.‡
MIMFormer	56.96	54.18	53.61	50.33	48.98	54.23	50.60	38.00	49.63	47.05	48.66	45.24	44.15	46.89	43.83	29.86
MIMFormer, randomized	56.50	56.38	55.66	51.67	51.30	56.43	54.90	37.46	49.37	49.28	48.31	45.87	45.22	49.32	48.21	29.33
ExactFuse	**58.05**	**57.79**	55.13	47.60	46.71	50.79	44.82	**38.38**	**50.63**	**50.50**	49.02	43.15	42.32	42.37	37.40	**30.31**
base model, randomized data	57.43	57.33	57.35	52.35	52.30	**57.18**	55.40	37.48	50.40	50.33	50.32	46.40	46.35	**50.01**	47.79	29.50
+ estimated operator, conditioning	57.61	57.51	57.54	**52.92**	**52.83**	54.78	50.87	38.12	50.42	50.35	50.38	**46.84**	**46.80**	47.02	43.20	30.01
ExactFuse, randomized + conditioning	57.77	57.68	**57.73**	52.64	52.59	55.34	51.90	38.06	50.43	50.38	**50.39**	46.55	46.53	46.76	43.01	30.06
wide estimate	57.77	57.68	**57.73**	52.64	52.59	56.84	56.07	37.48	50.43	50.38	**50.39**	46.55	46.53	48.71	49.12	29.32
ExactFuse, wider family (+ aniso, offsets)	57.75	57.65	57.70	52.55	52.50	56.85	**56.10**	37.48	50.40	50.36	50.36	46.55	46.54	48.84	**49.17**	29.32
conditioning, true operator	57.77	57.68	57.73	52.64	52.59	57.59	56.08	–	50.43	50.38	50.39	46.55	46.53	50.33	49.23	–

errors are 0.05 px in blur width and 0.0 nm in spectral shift. Training over the family makes the network robust to noise and combined changes: at 40–30 dB SNR the mean PSNR of ExactFuse rises from 47.60 to 52.64 dB on Chikusei, against 51.67 dB for MIMFormer trained on the same randomized data. In the base model, the estimated operator adds to randomized training alone (e.g., 52.83 against 52.30 dB under combined changes), and MIMFormer stays below both. The price is a lower PSNR at the training operator (57.77 against 58.05 dB). One network trained on random responses also serves seven real sensors it never saw, 3.2–11.3 dB above MIMFormer trained the same way on Chikusei and 1.79 dB below ExactFuse trained for WorldView-2, and one network trained at $\times 4$ and $\times 8$ leads the best retrained network at the unseen $\times 16$ by 1.76 dB (the supplementary material).

Outside the family the operator-conditioned network is fragile: with an anisotropic or defocus blur its estimated isotropic operator is wrong, and it drops to 55.34 dB on Chikusei, below the base model trained on randomized data with the nominal operator (57.18 dB). Estimating the wider separable family of (7), with two blur widths and sub-pixel grid offsets, gives 56.84 dB under these blurs and 56.07 dB under grid offsets, against 57.59 and 56.08 dB with the true operator. An HR-MSI misregistered by half a pixel costs

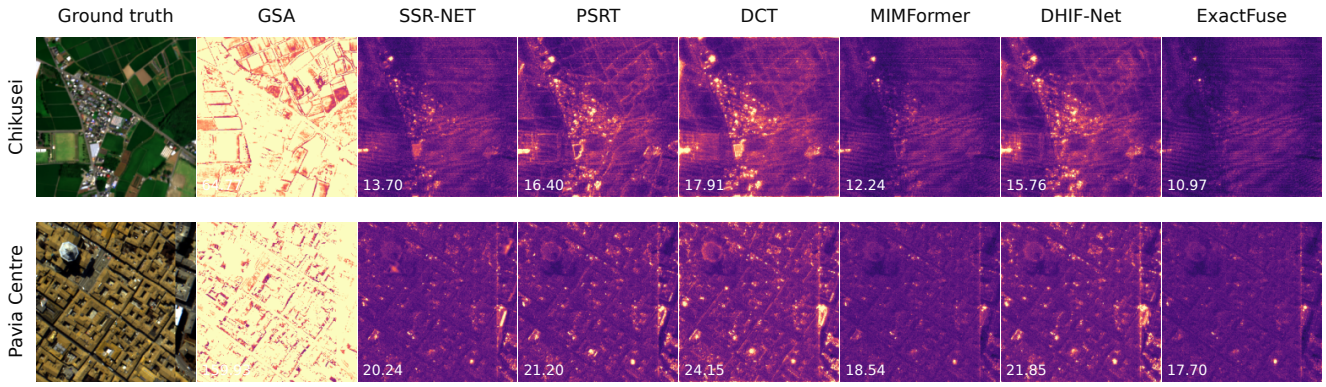


Figure 3. Mean absolute error over all bands (DN) on the test image where ExactFuse has its lowest PSNR. The color scale is set by the retrained networks (clipped at the 99.5th percentile), so GSA saturates; the number in each map is its mean.

Table 3. PSNR (dB) on CAVE and Harvard (unseen test scenes) and at larger scale factors, same protocol and budget as Tab. 1. –: not run (SMGU-Net's code supports ×4 only).

Method	CAVE ×4	Harv. ×4	CAVE ×8	Chik. ×8	Pavia ×8	Chik. ×16
GSA	42.30	46.30	38.01	38.69	30.24	35.20
SSR-NET	46.51	49.18	44.25	54.57	47.71	53.98
PSRT	48.51	49.29	45.21	53.05	47.13	–
DCT	46.36	48.92	43.17	51.77	45.51	–
MIMFormer	50.93	49.72	48.10	55.57	49.09	54.21
DHIF-Net	49.82	49.58	–	–	–	–
U2Net	48.04	48.70	–	–	–	–
MIMO-SST	50.39	49.83	–	55.88	48.90	–
OTPNet	–	–	–	56.11	49.51	–
SMGU-Net	48.54	49.34	–	–	–	–
CaFormer	47.91	49.09	–	–	–	–
ExactFuse	**52.46**	**50.03**	**50.39**	**56.95**	**50.05**	**56.11**

Table 4. Ablation of ExactFuse, each variant trained from scratch for 4 h: one change per row, then one input removed, and the base model for reference (all variants in the supplementary material). Cons.: PSNR$(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{H}})$. Swap: worst PSNR over test blurs $\sigma \in \{0.75, 1.25, 1.5\}\sigma_0$ when the true blur is given to the data steps (n/a: no operator to replace; –: not evaluated).

	Chikusei			Pavia Centre		
Variant	PSNR	Cons.	Swap	PSNR	Cons.	Swap
ExactFuse	58.05	93.4	57.93	50.63	82.5	50.57
gradient step instead of closed form	58.03	86.7	57.84	50.63	79.5	50.56
no data steps	57.42	67.5	n/a	50.30	68.1	n/a
+ null-space gate	57.97	96.9	57.82	50.58	87.0	50.48
+ gate, EON prior instead of U-Net	57.83	96.5	57.74	50.37	84.9	50.32
base model (gradient step, $\mathbf{Y}_{\mathrm{M}}$ input)	57.95	83.9	57.70	50.63	77.6	50.50

every method, GSA included, more than 15 dB. Registering the HR-MSI before fusion with the offset estimated from the pair recovers most of this loss: under a 0.5–1 px misregistration, ExactFuse rises from 37.88 to 49.74 dB on Chikusei (MIMFormer registered the same way 49.38 dB, GSA 42.42 dB). With an anisotropic blur and 35 dB noise added, ExactFuse trained on noise-free data falls behind MIMFormer (43.48 against 45.49 dB); trained over the wider family, it reaches 45.72 dB. Exactness thus makes the network precise when the operator family covers the test conditions and fragile when it does not.

6. Limitations and conclusion

Limitations. (i) Results with a reference rest on simulated observations (Wald's protocol), as in every benchmark of this field; on real NEON data (spectrometer and RGB camera, the supplementary material), ExactFuse is again the most accurate at reduced resolution (38.70 against 38.45 dB for MIMFormer), but its operator variants differ by less than 0.1 dB there, because the fitted camera response is only approximate. (ii) Exact operators help only when the operator family contains the true operator: rotated anisotropic and defocus blurs fall outside it (Tab. 2), registration leaves several decibels of a misregistration loss, and non-separable kernels lose the closed form. (iii) All methods train for 4 h; 8 h narrows the gap but keeps it, and converged comparisons were not run. (iv) The Chikusei and Pavia Centre test sets are small; three seeds, the pooled test and CAVE and Harvard back the comparison. (v) The eigendecompositions grow cubically with the image side and are cached per image size.

Exact separable operators with the true border give unfolded fusion a closed-form data step, an explicit null space, a bounded effect of operator errors and operators identifiable from the two observations. Under one protocol and budget, ExactFuse is more accurate than eleven retrained networks, mainly through its prior and training, and its exact operators let one network adapt to new blurs, responses, noise, sensors and scale factors. Spatially varying operators, band-dependent noise and real paired references remain open. Code will be made publicly available.

LLM Usage Statement

A large language model assistant was used extensively in this work: to write and debug the training and evaluation code, to run and analyze the experiments on cloud GPUs, and to draft and edit the text. The authors directed the work, reviewed the code, results and claims, and take full responsibility for the content.

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